

Chapter 1

how do these papers handle attention mechanisms

Attention mechanisms emerge across these papers as a solution to the representational bottleneck of compressing variable-length sequences into fixed-size vectors, with later work generalizing attention from an auxiliary alignment tool into the sole computational primitive of the architecture. [5] exemplifies the problem attention was designed to solve: their sequence-to-sequence LSTM encodes an entire source sentence into one fixed-dimensional vector, a design that works only because of an ad hoc trick reversing the source sentence to shorten effective dependency distances, and the paper explicitly notes this fixed-vector bottleneck as a limitation motivating subsequent work [5]. [2] directly addresses this bottleneck by introducing a soft-alignment "attention" mechanism that lets the decoder compute a weighted context vector over all encoder hidden states at each output step, rather than relying on a single fixed-length summary; this frees the encoder from having to compress all sentence information into one vector and yields the largest gains on long sentences, precisely the cases where the fixed-vector approach of [5] struggles. [6] then takes this alignment idea to its logical extreme, dispensing with recurrence and convolution entirely and building a model, the Transformer, based solely on scaled dot-product and multi-head self-attention; where [2] uses attention as an add-on to an RNN encoder-decoder, [6] uses self-attention as the mechanism by which every position in a sequence directly attends to every other position, achieving constant maximum path length between any two positions and much greater parallelism. [3] builds directly on this architecture, adopting the Transformer's bidirectional self-attention as BERT's core encoder and contrasting it explicitly with the constrained, left-to-right self-attention used in GPT-style unidirectional Transformers, showing that allowing every token to attend freely in both directions (via the masked-language-model objective) yields substantially better representations for downstream token- and sentence-level tasks [3].

[1] and [4] do not address attention mechanisms in this sense: the former proposes layer-wise relevance propagation for explaining classifier decisions in feedforward and Bag-of-Words models, and the latter introduces residual connections for training very deep image classification networks, so both are omitted from the comparison above.

Bibliography

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